The identity type $\mathsf{Id}\,A\,a\,b$ with Ref and J

typing, reduction, injectivity, and subject reduction

Agda development `ID/` (domain-semantics)

This note records the rules for the element-level Martin-Löf identity type added to the type-in-type theory ($\mathsf{U} : \mathsf{U}$) of the `ID/` development, and states the two metatheorems proved for it. Everything below is machine-checked in Agda (`--without-K --exact-split`) with no postulates, no holes, and no termination/positivity pragmas.

The eliminator is Martin-Löf's *original* J with a fully dependent motive. Writing the motive and base types out,

$$C : (x\,y : A) \to \mathsf{Id}\,A\,x\,y \to \mathsf{U}, \qquad d : (x : A) \to C\,x\,x\,(\mathsf{Ref}\,x),$$

so that $\mathsf{J}\,C\,d\,p$ at $p : \mathsf{Id}\,A\,a\,b$ has type $C\,a\,b\,p$ (the application spine $((C\,a)\,b)\,p$).

Typing rules

$$\textsc{Id}\quad \frac{\Gamma \vdash A : \mathsf{U} \qquad \Gamma \vdash a : A \qquad \Gamma \vdash b : A}{\Gamma \vdash \mathsf{Id}\,A\,a\,b : \mathsf{U}} \qquad\qquad \textsc{Ref}\quad \frac{\Gamma \vdash A : \mathsf{U} \qquad \Gamma \vdash a : A}{\Gamma \vdash \mathsf{Ref}\,a : \mathsf{Id}\,A\,a\,a}$$

$$\textsc{J}\quad \frac{\begin{array}{c}\Gamma \vdash A : \mathsf{U} \qquad \Gamma \vdash a : A \qquad \Gamma \vdash b : A \\ \Gamma \vdash C : (x\,y : A) \to \mathsf{Id}\,A\,x\,y \to \mathsf{U} \qquad \Gamma \vdash d : (x : A) \to C\,x\,x\,(\mathsf{Ref}\,x) \qquad \Gamma \vdash p : \mathsf{Id}\,A\,a\,b\end{array}}{\Gamma \vdash \mathsf{J}\,C\,d\,p : C\,a\,b\,p}$$

Conversion rules

$$\textsc{J-}\beta\quad \frac{\Gamma \vdash A : \mathsf{U} \qquad \Gamma \vdash a_0 : A \qquad \Gamma \vdash C : \ldots \qquad \Gamma \vdash d : \ldots}{\Gamma \vdash \mathsf{J}\,C\,d\,(\mathsf{Ref}\,a_0) = d\,a_0 : C\,a_0\,a_0\,(\mathsf{Ref}\,a_0)}$$

J fires only on a *literal* $\mathsf{Ref}\,a_0$, whose type is the diagonal $\mathsf{Id}\,A\,a_0\,a_0$; both sides of J-β therefore have the *same* type $C\,a_0\,a_0\,(\mathsf{Ref}\,a_0)$, so the rule carries no endpoint or motive equalities and needs no injectivity. The congruences are

$$\textsc{Id-cong}\quad \frac{\Gamma \vdash A = A' : \mathsf{U} \qquad \Gamma \vdash a = a' : A \qquad \Gamma \vdash b = b' : A}{\Gamma \vdash \mathsf{Id}\,A\,a\,b = \mathsf{Id}\,A'\,a'\,b' : \mathsf{U}} \qquad\qquad \textsc{Ref-cong}\quad \frac{\Gamma \vdash a = a' : A}{\Gamma \vdash \mathsf{Ref}\,a = \mathsf{Ref}\,a' : \mathsf{Id}\,A\,a\,a}$$

$$\textsc{J-cong}\quad \frac{\Gamma \vdash C = C' : \ldots \qquad \Gamma \vdash d = d' : \ldots \qquad \Gamma \vdash p = p' : \mathsf{Id}\,A\,a\,b}{\Gamma \vdash \mathsf{J}\,C\,d\,p = \mathsf{J}\,C'\,d'\,p' : C\,a\,b\,p}$$

(Grey premises re-typing the components are elided; see `ID/Syntax/Typing.agda`.)

Head reduction

Reduction is untyped weak head reduction. The single-step relation $M \rightsquigarrow_1 N$ has the following Id-clauses (alongside the usual β and application-congruence for functions):

$$\frac{\textsc{J-red}}{\mathsf{J}\, C\, d\, (\mathsf{Ref}\, a) \ \rightsquigarrow_1\ d\, a} \qquad\qquad \frac{\textsc{J-scrut}\quad p \rightsquigarrow_1 p'}{\mathsf{J}\, C\, d\, p \ \rightsquigarrow_1\ \mathsf{J}\, C\, d\, p'}$$

J-red is the ι-step; J-scrut lets reduction proceed into the proof position so a non-canonical proof can reduce toward a Ref. Each step is a typed conversion (J-red $\mapsto$ J-β, J-scrut $\mapsto$ J-cong), and $\rightsquigarrow_1$ is deterministic.

Metatheorems

Theorem 1 (Id-injectivity, `ID/IdInjectivity.agda`). *If* $\Gamma \vdash \mathsf{Id}\, A_0\, a_0\, b_0 = \mathsf{Id}\, A_1\, a_1\, b_1 : \mathsf{U}$ *then*

$$\Gamma \vdash A_0 = A_1 : \mathsf{U}, \qquad \Gamma \vdash a_0 = a_1 : A_0, \qquad \Gamma \vdash b_0 = b_1 : A_0.$$

As with Π-injectivity, this is read off adequacy: evaluate the given conversion at the bottom environment, obtaining an equality in the model at the type-code $\mathsf{IdCode}\, \bot\, \bot\, \bot$, and project out the component conversions (lemma `idConv` + `HeadRed-unique-Id`).

Theorem 2 (Subject reduction, `ID/SubjectReduction.agda`). *If* $\Gamma \vdash M : A$ *and* $M \rightsquigarrow_1 N$ *then* $\Gamma \vdash N : A$.

For the J-red step the contractum $d\, a_0$ already has type $C\, a_0\, a_0\, (\mathsf{Ref}\, a_0)$, which is retyped to the goal $C\, a\, b\, (\mathsf{Ref}\, a_0)$ using $a_0 = a$ and $a_0 = b$ extracted from $\mathsf{Ref}\, a_0 : \mathsf{Id}\, A\, a\, b$ (`ty-Ref-endpoints`). For the J-scrut step the result type $C\, a\, b\, p$ is transported along $p = p'$, obtained from *reduction-is-conversion* (`headred1-conv`); the Id-typed proof never needs Id-injectivity, only the diagonal J-β.

Both theorems sit on top of adequacy (`ID/Adequacy/Value.agda`: `adequacySub2` / `adequacyEqSub2`), whose J case is the based-J driver (`ID/Adequacy/JApp.agda`, `JAppCross.agda`, `JAppCongr.agda`).